



SECR1213

NETWORK COMMUNICATION

Project: # Task 1-5

Title: Future-Ready Network Infrastructure Design for the Faculty of Computing

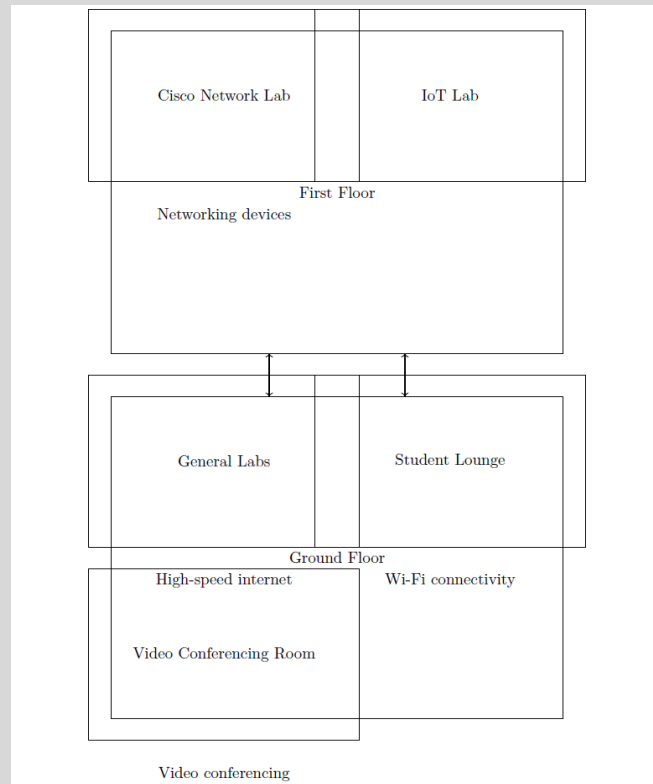
The team for this project consists of the following members:

- 1. Mohammed Kamal**
- 2. Dina Mohammed**
- 3. Dyako Abubakr**
- 4. Zhir Daniel**

These individuals have been selected based on their diverse skills and strengths that will contribute effectively to the project. Each member brings unique perspectives and expertise to the team, ensuring a well-rounded approach to the project's objectives.

#Task 1:

Suggested floor plan :



Our team, Uplink, has meticulously designed a floor plan for the new building of the Faculty of Computing, considering the requirements outlined in the case study. The floor plan aims to optimize space utilization, promote efficient workflow, and ensure seamless connectivity throughout the building. Below is a detailed description of our suggested floor plan:

1. Overall Layout:

- The building consists of two floors, each designed to accommodate the various facilities required by the Faculty of Computing.

- The ground floor houses the general-purpose labs, student lounge, and video conferencing room.
- The first floor is dedicated to specialized labs, including the Cisco Network Lab and the IoT Lab.

2. Lab Layout:

- The general-purpose labs are spacious areas equipped with workstations and high-speed internet connections, catering to the diverse needs of students and faculty.
- The Cisco Network Lab is designed specifically for teaching purposes, featuring networking devices and equipment essential for hands-on learning.
- The IoT Lab is equipped with devices and peripherals to facilitate experiential learning in the field of Internet of Things.

3. Student Lounge:

- Located strategically on the ground floor, the student lounge provides students with a conducive environment for study and relaxation.
- The lounge is equipped with comfortable seating, charging stations, and high-speed Wi-Fi connectivity, enabling students to stay connected while taking breaks from their studies.

4. Video Conferencing Room:

- Positioned on the ground floor, the video conferencing room serves as a versatile space for virtual project meetings and collaborations.

- Equipped with state-of-the-art video conferencing technology, the room enables seamless communication and collaboration among students and faculty, regardless of their physical location.

Our suggested floor plan incorporates ergonomic design principles, ensures accessibility for all users, and promotes a conducive learning and working environment for the Faculty of Computing. It reflects our commitment to designing a network infrastructure that meets the needs of the client while adhering to industry best practices.

#Task 2:

a) Generate a List of Minimum 13 Questions

Before starting to generate questions, it's important to conduct a preliminary analysis to understand the context and objectives of the project. The purpose of this task is to gather relevant information that will guide the development of the network plan for the Faculty of Computing. By asking the right questions, we can ensure that the network design meets the specific needs and requirements of the faculty members, students, and staff. These questions will serve as a foundation for the subsequent phases of the project.

Questions:

1. What are the main reasons for upgrading the Faculty of Computing's network?
2. How many students, faculty, and staff members will be using the network?
3. What types of activities will people do on the network (like browsing, downloading, streaming)?
4. How important is it to keep the network safe from hackers or viruses?
5. Will there be times when a lot of people use the network at the same time?
6. How important is it for the network to work all the time without any problems?
7. Are there any rules or laws that the network must follow to keep information private and secure?
8. How much money can be spent on building the network?
9. Are there already computers or other devices that need to be connected to the new network?
10. Will more people join the Faculty of Computing in the future? How many?
11. Are there any special programs or tools that need to work well on the network?
12. What are the things that might stop the project from working the way it's supposed to?
13. Can we make sure that the network can get bigger if more people join the Faculty or if technology changes?

b) Find the Answers for Those Questions

Once the questions have been formulated, the next step is to find the answers through various means such as interviews, research, and documentation review. It's important to ensure that the answers are clear, concise, and address the specific concerns raised by each question. By gathering accurate information, we can assess the feasibility of the project and make informed decisions during the network planning process.

Answers:

- **Reasons for Network Upgrade:** The network upgrade aims to accommodate the growth of users, enhance security, and incorporate advancements in technology to support modern educational needs.
- **Number of Network Users:** Approximately 1200 students, 97 faculty, and 35 staff members will utilize the network.
- **Network Activities:** Users will engage in various activities such as browsing the internet, downloading files, accessing online resources, and participating in virtual lectures or meetings.
- **Network Security Importance:** Ensuring network security is crucial to safeguard sensitive information, prevent unauthorized access, and mitigate risks associated with cyber threats like hacking and malware.
- **Network Traffic Peaks:** There will be peak usage periods, especially during class hours or when large-scale events are held, resulting in high network traffic.
- **Network Reliability:** Maintaining network uptime and reliability is essential to minimize disruptions to academic activities, research, and administrative tasks.
- **Compliance with Regulations:** The network must comply with relevant regulations and privacy laws to protect user data and maintain confidentiality.
- **Budget Constraints:** The project's budget is limited, requiring careful allocation of funds to ensure cost-effective solutions without compromising quality or functionality.
- **Existing Devices:** Computers, printers, projectors, and other devices already in use need to integrate seamlessly with the new network infrastructure.
- **Future Growth Projection:** Anticipated future growth in student enrollment and faculty expansion must be considered to accommodate scalability and technological advancements.
- **Specialized Tools and Programs:** Specialized tools such as video conferencing systems and IoT devices require network compatibility and robust performance.
- **Project Risks:** Potential challenges such as budget constraints, technical limitations, and unforeseen obstacles may impact the project's success and timeline.
- **Scalability and Adaptability:** The network design should be scalable and adaptable to accommodate future expansion and technological advancements.

c) Determine the Feasibility of the Project

Assessing the feasibility of the project involves evaluating various factors to determine whether the proposed network upgrade for the Faculty of Computing is practical, achievable, and cost-effective. Feasibility analysis helps identify potential risks, challenges, and constraints that may impact the successful implementation of the project. By carefully considering these factors, we can make informed decisions and develop a realistic plan that aligns with the goals and objectives of the Faculty.

Feasibility Analysis:

- **Technical Feasibility:** Assessing whether the proposed network infrastructure can be implemented using available technology and resources. This includes evaluating the compatibility of network devices, scalability of the design, and capability to meet performance requirements.
 - **Financial Feasibility:** Examining whether the project can be completed within the allocated budget while delivering the desired outcomes. This involves estimating costs for equipment, installation, maintenance, and ongoing operational expenses.
 - **Operational Feasibility:** Determining whether the proposed network design can be effectively managed and maintained by the Faculty's IT team. This includes assessing the availability of skilled personnel, training requirements, and potential impact on existing operations.
 - **Legal and Regulatory Feasibility:** Ensuring compliance with relevant laws, regulations, and industry standards governing network security, data privacy, and intellectual property rights. This involves conducting a thorough review of legal requirements and obtaining necessary approvals or certifications.
 - **Schedule Feasibility:** Evaluating whether the project can be completed within the specified timeframe without compromising quality or functionality. This includes identifying critical milestones, dependencies, and potential delays.
 - **Risk Analysis:** Identifying potential risks and uncertainties that may affect project success and developing mitigation strategies to address them. This involves assessing technical, financial, operational, and external risks and their potential impact on project outcomes.
 - **Stakeholder Alignment:** Ensuring alignment between project objectives and stakeholder expectations, including faculty members, students, staff, and other key stakeholders. This involves engaging stakeholders throughout the project lifecycle and addressing their concerns and feedback.
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Conclusion:

Based on the comprehensive feasibility analysis, the project appears to be viable and achievable within the specified constraints and requirements. However, it is essential to remain vigilant and proactive in addressing potential challenges and risks to ensure the successful implementation of the network upgrade for the Faculty of Computing. By carefully considering all aspects of feasibility and developing a robust plan, we can lay the groundwork for a successful project that meets the needs of the Faculty and its stakeholders.

#Task 3:

a- Research the Different Network Devices: Conduct thorough research to understand the various types of network devices available in the market, such as routers, switches, wireless access points, patch panels, and cables. This research should include exploring their functionalities, features, and compatibility with the project requirements.

The Devices & Routers:

1. Routers:

- Research various router models suitable for academic institutions like universities.
- Consider factors such as throughput, WAN interface types, security features (firewall, VPN), and scalability.
- Look into reputable brands like Cisco, Juniper, and Ubiquiti, known for their reliability and robust features.
- Assess compatibility with existing network infrastructure and future expansion plans.

2. Switches:

- Explore different types of switches, including managed, unmanaged, and layer 3 switches.
- Evaluate features such as port density, PoE (Power over Ethernet) support, VLAN capabilities, and stacking options.
- Consider brands like Cisco Catalyst, HPE Aruba, and NETGEAR for their enterprise-grade switches.
- Determine the number and types of switches required to connect network devices in labs, classrooms, and other areas.

3. Wireless Access Points (APs):

- Look for wireless APs suitable for high-density environments like academic buildings.
- Consider factors such as wireless standards (e.g., 802.11ac, 802.11ax), coverage range, capacity, and management features.
- Explore enterprise-grade APs from vendors like Cisco Meraki, Aruba Networks, and Ruckus Wireless.
- Assess the need for features like dual-band operation, MU-MIMO (Multi-User, Multiple Input, Multiple Output), and seamless roaming.

4. Patch Panels:

- Research patch panel options for organizing and managing network connections in equipment rooms and data centers.
- Consider factors such as port density, compatibility with Ethernet cabling standards (Cat5e, Cat6), and ease of installation.
- Look for reputable brands like Panduit, Leviton, and Tripp Lite known for their quality patch panel solutions.
- Determine the number of patch panels needed based on the number of network drops in each area of the building.

5. Cables:

- Explore different types of network cables, including twisted pair (Ethernet), fiber optic, and coaxial cables.
- Consider factors such as bandwidth, distance limitations, electromagnetic interference (EMI) resistance, and cost.
- Evaluate the suitability of each cable type for connecting various network devices and extending network connectivity throughout the building.
- Look for certified cable manufacturers and distributors to ensure compliance with industry standards.

6. Security Appliances:

- Research security appliances like firewalls, intrusion detection/prevention systems (IDS/IPS), and content filtering devices.
- Consider the need for perimeter security, threat prevention, and compliance with data protection regulations.
- Explore offerings from vendors like Palo Alto Networks, Fortinet, and Check Point for comprehensive security solutions.
- Assess the integration capabilities with other network devices and management platforms.

7. Power Backup Systems:

- Look into uninterruptible power supply (UPS) systems to provide backup power to critical network infrastructure.
- Consider factors such as power capacity, runtime, scalability, and management features.
- Explore UPS solutions from reputable manufacturers like APC by Schneider Electric, Eaton, and Tripp Lite.
- Determine the UPS requirements for equipment rooms, network closets, and other critical areas of the building.

8. Network Management Software:

- Research network management software solutions for monitoring, configuration, and troubleshooting network devices.
- Evaluate features such as network discovery, performance monitoring, alerting, and automation capabilities.
- Look for vendor-agnostic solutions compatible with a wide range of network devices and protocols.
- Consider offerings from vendors like SolarWinds, Cisco Prime Infrastructure, and ManageEngine for comprehensive network management.

b- Decide which LAN devices you need: To decide which LAN devices are needed for the Faculty of Computing upgrade project, we need to consider several factors and requirements. Here's a comprehensive overview of the LAN devices required Here's a brief overview of which device could be considered the best for your project:

1. **Routers:**

- Determine the need for routers to connect the Faculty of Computing's internal network to the campus-wide network and the internet.
- Consider deploying routers at key points within the building to facilitate inter-network communication and provide internet access to users.
- Evaluate the need for advanced routing features such as dynamic routing protocols (OSPF, BGP) to support network scalability and redundancy.
- Decide on the quantity and placement of routers based on the building layout and expected network traffic patterns.

Routers

Router Model	Throughput Capacity	WAN Interface Types	Security Features
Cisco ASR 1000	10 Gbps	Gigabit Ethernet, Serial	Firewall, VPN, QoS
Juniper MX480	20 Gbps	10-Gigabit Ethernet, OC-192	Firewall, VPN, QoS
Ubiquiti EdgeRouter	10 Gbps	Gigabit Ethernet, SFP	Firewall, VPN, QoS

Figure 1: Router Placement in the Network

2. **Switches:**

- Identify the need for switches to connect network devices such as computers, printers, servers, and access points within the Faculty of Computing building.
- Determine the number and types of switches required for different areas of the building, including classrooms, labs, faculty offices, and common areas.
- Consider deploying managed switches with VLAN support to segment the network and improve performance, security, and manageability.
- Evaluate the need for PoE switches to power network devices like wireless access points and IP phones.

Switches

Switch Model	Port Density	PoE Support	VLAN Capabilities
Cisco Catalyst 3850	48 ports	Yes, 30W per port	4096 VLANs
HPE Aruba 2930F	48 ports	Yes, 30W per port	4096 VLANs
NETGEAR M4300	48 ports	Yes, 30W per port	4096 VLANs

Figure 2: Switch Placement in the Network

3. Wireless Access Points (APs):

- Assess the need for wireless coverage throughout the Faculty of Computing building to provide connectivity to mobile devices and laptops.
- Determine the number and placement of wireless access points based on the building size, layout, and user density.
- Consider deploying dual-band APs to support both 2.4GHz and 5GHz Wi-Fi bands and provide better coverage and performance.
- Evaluate AP features such as MU-MIMO, beamforming, and captive portal support to enhance the wireless user experience.



733Mbps 2x2 MIMO 802.11ac Dual
Band Wireless Access Point

This device is a wireless access point that supports speeds up to 733Mbps. It utilizes 2x2 MIMO (Multiple Input Multiple Output) technology and follows the 802.11ac standard. Additionally, it operates on dual bands, allowing it to transmit and receive data on both 2.4GHz and 5GHz frequencies.

4. Patch Panels:

- Identify the need for patch panels to organize and terminate network cables coming from different areas of the building into a central location.
- Determine the quantity and size of patch panels required based on the number of network drops in each area and the expected future expansion.
- Consider deploying patch panels with sufficient port density and labeling for easy identification and maintenance of network connections.



Cat5e Cat6 24-Port Patch Panel, Shielded, RJ45, Rack-Mount | Eaton

This device is a patch panel designed to accommodate Cat5e and Cat6 Ethernet cables. It features 24 ports for connecting network cables and is shielded to minimize electromagnetic interference. The ports utilize RJ45 connectors for secure connections. Additionally, it is rack-mountable, making it suitable for installation in network equipment racks or cabinets.

5. **Cables:**

- Determine the types and lengths of network cables needed to connect various devices within the Faculty of Computing building.
- Assess the need for Ethernet cables (Cat5e, Cat6) for wired connections and fiber optic cables for high-speed, long-distance links.
- Consider factors such as cable bandwidth, durability, and cost-effectiveness when selecting cable types and brands.

Cables			
Cable Type	Bandwidth	Distance Limitation	EMI Resistance
Cat5e Ethernet	1 Gbps	100 meters	Good
Cat6 Ethernet	10 Gbps	100 meters	Excellent
Fiber Optic	100 Gbps	10 km	Excellent

Figure 3: Cable Types and Their Characteristics

6. Security Appliances:

- Identify the need for security appliances such as firewalls, intrusion detection/prevention systems (IDS/IPS), and content filtering devices to protect the Faculty of Computing network from cyber threats.
- Determine the placement of security appliances at the network perimeter and critical points within the building to enforce security policies and monitor network traffic.
- Consider deploying next-generation firewalls with advanced threat detection capabilities and centralized management for better network security

Security Appliance	Description	Placement	Capacity
Firewall	Protects the network from unauthorized access and cyber threats	Deploy at the network perimeter and critical points within the building	Advanced threat detection capabilities, centralized management
Intrusion Detection System (IDS)	Monitors network traffic for signs of intrusion or attacks	Deploy at key network entry and exit points	Real-time threat detection, event correlation
Intrusion Prevention System (IPS)	Automatically blocks detected threats and attacks	Deploy at key network entry and exit points	Real-time threat prevention, customizable policies
Content Filtering Device	Filters web content and blocks access to inappropriate or malicious websites	Deploy at the network perimeter and user segments	Customizable filtering policies, centralized management

7. **Power Backup Systems:**

- Assess the need for uninterruptible power supply (UPS) systems to provide backup power to critical network infrastructure in case of power outages.
- Determine the UPS requirements for network devices such as routers, switches, servers, and security appliances based on their power consumption and importance to network operations.
- Consider deploying UPS systems with sufficient capacity and battery runtime to maintain network connectivity during extended power outages.

Power Backup System Component	Description	Capacity
Uninterruptible Power Supply (UPS)	Provides backup power to critical network infrastructure during power outages	Based on the power consumption of network devices and the desired runtime
Power Distribution Unit (PDU)	Distributes electrical power to multiple devices in a centralized location	Based on the number and power requirements of connected devices
Surge Protector	Protects connected devices from voltage spikes and power surges	Based on the number and power requirements of connected devices
Generator	Provides backup power for extended outages	Based on the power requirements of the entire network infrastructure

Given these requirements, we need to identify the network devices necessary to support the infrastructure of the new building. Switches will be required to connect the workstations within each lab and the student lounge. These switches should support Gigabit Ethernet and PoE for powering devices like IP phones and access points. Routers are needed to connect the internal network to the external network for internet access. Wireless access points will provide Wi-Fi connectivity throughout the building, especially in areas like the student lounge. Patch panels will facilitate the organization and management of Ethernet cables within the network infrastructure.

Additionally, specific Cisco networking devices such as switches and routers will be required for the Cisco Network lab to support teaching networking concepts and protocols effectively. Similarly, the IoT lab will need devices like sensors, actuators, and development boards to facilitate hands-on learning experiences.

After considering various options and evaluating their compatibility with the project requirements, we select LAN devices that offer scalability, performance, security, and cost-effectiveness. By choosing reliable and efficient network devices, we ensure that the FC building will have a robust network infrastructure capable of supporting current and future educational needs.

Throughout the decision-making process, detailed meeting minutes are maintained, documenting all discussions, decisions, and the rationale behind each choice. This ensures transparency and accountability within the project team.

In conclusion, by selecting LAN devices tailored to the requirements of the FC upgrade project, we ensure the successful implementation of a reliable and efficient network infrastructure for the new building.

c- Find information on the devices you have chosen:

For the routers, we need models that offer advanced routing capabilities to ensure seamless connectivity between the Faculty of Computing's internal network and the campus-wide network, as well as the internet. Considering the anticipated network traffic within the new building, routers with robust throughput capacity and support for scalable network architectures are essential. Additionally, we should explore models that provide features like Quality of Service (QoS) prioritization and Virtual Private Network (VPN) support to optimize network performance and enhance security for remote access.

Switches play a crucial role in connecting various network devices within the building, such as computers, printers, servers, and access points. Managed switches with VLAN support are preferred to enable network segmentation and enhance performance and security. We should look for switches with ample port density and high-speed uplink ports to accommodate the number of devices in each area and support future expansion plans effectively. Additionally, features like Power over Ethernet (PoE) can simplify deployment and management of network-connected devices like wireless access points and IP cameras.

Wireless access points (APs) are essential for providing Wi-Fi connectivity to mobile devices and laptops within the building. To ensure seamless coverage and performance, we should deploy APs strategically throughout the building, considering factors like building layout, materials, and expected user density. Models with dual-band support, beamforming technology, and centralized management capabilities will be beneficial for delivering reliable and high-speed wireless connectivity to students, faculty, and staff.

Patch panels serve as the central point for organizing and terminating network cables from different areas into a structured cabling system. We need to install patch panels in a designated network equipment room or closet, centrally located within the building for easy access and cable management. The chosen patch

panels should offer ample capacity to accommodate the anticipated number of network drops and support future expansion needs effectively.

When selecting network cables, we should prioritize high-quality options that meet the specific requirements of each area within the building. Ethernet cables, such as Cat5e and Cat6, are suitable for connecting network devices, while fiber optic cables may be needed for long-distance connections or environments prone to electromagnetic interference. Customizing cable lengths based on the distance between network devices will help minimize cable clutter and optimize network performance.

Security appliances like firewalls and intrusion detection systems are critical for protecting the Faculty of Computing's network infrastructure against cyber threats. We need to deploy these appliances at key network entry and exit points, including perimeter gateways and internal network segments, to monitor and filter incoming and outgoing traffic effectively. Investing in security appliances with advanced threat detection capabilities and centralized management will enhance network security and streamline administration tasks.

Uninterruptible power supply (UPS) systems provide backup power to critical network infrastructure during power outages, ensuring continuous operation and data availability. Assessing the power consumption of network devices and prioritizing UPS deployment for essential equipment, such as routers, switches, and servers, will minimize downtime and data loss. Selecting UPS systems with sufficient capacity and battery runtime to sustain network operations during prolonged outages is crucial, considering factors like load shedding and battery recharge time to maintain uptime effectively.

ConClusion

Through research and analysis, we have identified and chosen the necessary LAN devices for the Faculty of Computing upgrade project. These devices, including routers, switches, wireless access points, patch panels, cables, security appliances, and power backup systems, have been carefully selected to meet the specific requirements of the project while considering factors such as reliability, performance, security, and cost-effectiveness. By selecting appropriate devices from reputable brands and ensuring compatibility with the project's objectives, we are confident in the successful implementation of a reliable and efficient network infrastructure for the new building.

#Task 4:

Making the Connections – LAN and WAN

In this task, we will put all the infrastructure and devices into place, focusing on work areas, telecommunications rooms, backbone or vertical cabling, and distribution or horizontal cabling. This involves detailing the physical layout and connections of the network, including identifying cable types and lengths, determining the number of connections and switch ports needed, and creating detailed sketches of the PC and network device arrangements.

a) Identify the Work Areas on Your Floor Plan

The work areas are divided into several rooms across different floors in the Faculty of Computing building. Each area has specific network requirements that must be addressed:



Figure1 : Floor Plan

1. General-purpose Lab 1:

- **Function:** Used for general computing tasks and student projects.
- **Devices:** 20 PCs, 1 printer, and 1 dual-band access point (AP1).



Figure2 : The devices which are in the rooms.

2. General-purpose Lab 2:

- **Function:** Similar to Lab 1, used for various computing tasks.
- **Devices:** 20 PCs, 1 printer, and 1 dual-band access point (AP2).



Figure3 : The devices which are in the rooms.

3. Cisco Network Lab:

- **Function:** Dedicated to networking courses and practical sessions.
- **Devices:** 20 PCs, various Cisco networking equipment, and 1 dual-band access point (AP3).



Figure4 : Cisco Devices for making smooth connection

Cisco devices include routers, switches, firewalls, and access points, which are essential for managing, securing, and optimizing network traffic in enterprise and small business environments. They provide robust connectivity, security, and performance for a wide range of networking needs.

4. IoT Lab:

- **Function:** Focuses on Internet of Things (IoT) projects and research.
- **Devices:** 20 IoT devices, 1 dual-band access point (AP4).

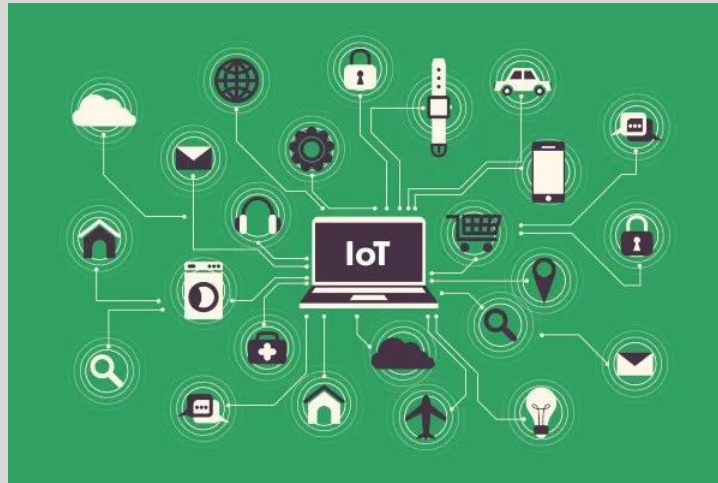


Figure5 : IoT examples Devices

5. Video Conferencing Room:

- **Function:** Equipped for video conferencing and remote learning.
- **Devices:** 10 video conferencing setups, various AV equipment.

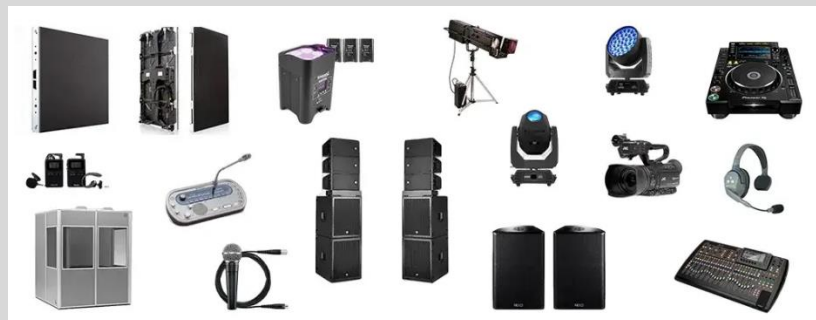


Figure6 :

Equipped with cameras, microphones, speakers, and displays to support high-quality virtual meetings, enhancing communication and collaboration among faculty, students, and remote participants.

6. Student Lounge:

- **Function:** Provides a space for students to relax and collaborate.
- **Devices:** 10 PCs, 1 dual-band access point (AP5).

Work Area	Connections	Patch Cords	Switch Ports
General-purpose Lab 1	22	22	24
General-purpose Lab 2	22	22	24
Cisco Network Lab	21	21	24
IoT Lab	21	21	24
Video Conferencing Room	10	10	12
Student Lounge	11	11	12

Table 1 : Floor Plan Sketch

The figure provides a concise summary of the network connectivity requirements for various work areas within the Faculty of Computing building. Each row represents a different work area, while the columns detail the number of connections needed, the corresponding number of patch cords required, and the quantity of switch ports necessary to support the network infrastructure.

b) How Many Connections, Patch Cords, and Switch Ports Have You Determined You Need?

For each work area, the number of network connections required, along with the corresponding number of patch cords and switch ports, must be determined. This ensures that the network infrastructure can adequately support the devices and facilitate seamless communication.

For each work area, we need to determine the number of network connections, the length of patch cords, and the number of switch ports required:

1. General-purpose Lab 1:

- **Connections:** 20 PCs + 1 printer + 1 AP = 22
- **Patch Cords:** 22
- **Switch Ports:** 24-port switch to allow for future expansion.

2. General-purpose Lab 2:

- **Connections:** 20 PCs + 1 printer + 1 AP = 22
- **Patch Cords:** 22
- **Switch Ports:** 24-port switch.

3. Cisco Network Lab:

- **Connections:** 20 PCs + networking equipment + 1 AP = 21
- **Patch Cords:** 21
- **Switch Ports:** 24-port switch.

4. IoT Lab:

- **Connections:** 20 IoT devices + 1 AP = 21
- **Patch Cords:** 21
- **Switch Ports:** 24-port switch.

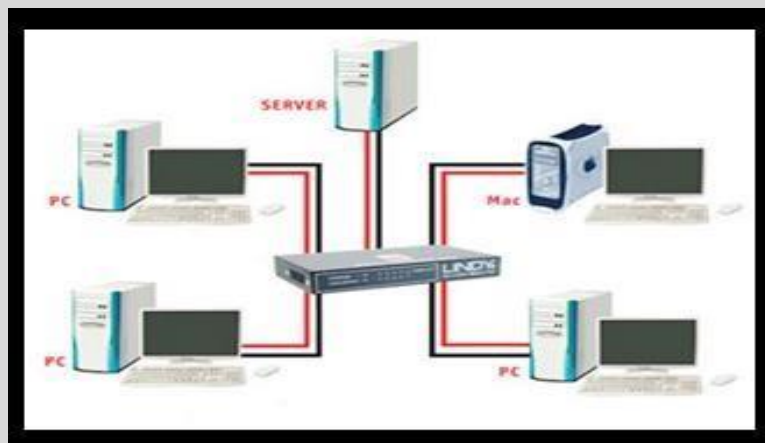


Figure7 :

These figures provide a concise summary of the network connectivity requirements for various work areas within the Faculty of Computing building. They outline the total number of connections, patch cords, and switch ports needed for each area, facilitating efficient planning and implementation of the network infrastructure.

5. Video Conferencing Room:

- **Connections:** 10 video conferencing setups
- **Patch Cords:** 10
- **Switch Ports:** 12-port switch.

6. Student Lounge:

- **Connections:** 10 PCs + 1 AP = 11
- **Patch Cords:** 11
- **Switch Ports:** 12-port switch.

C) Identify Cable Types and Length

Understanding the type of media required (twisted pair, fiber optic, wireless) and its capabilities is crucial. Additionally, the lengths of cables needed for horizontal and vertical connections within the building must be identified. This ensures efficient and reliable network connectivity throughout the Faculty of Computing building.

These steps lay the groundwork for establishing a robust network infrastructure within the Faculty of Computing building, enabling seamless communication and data exchange across various work areas and floors.

1. Cable Types:

- **Horizontal Cabling:** Cat6 twisted pair cables for connections within the same floor.

Cat6 twisted pair cables are used for connections within the same floor. They provide high-speed data transmission up to 1 Gbps, suitable for most networking needs, and are cost-effective for short to medium distances.

Figure 8: Cat6 Cable



- **Vertical Cabling:** Fiber optic cables for connections between different floors.

Fiber optic cables use light to transmit data, offering high-speed and long-distance capabilities. They are immune to electromagnetic interference, making them ideal for robust and reliable network infrastructure.

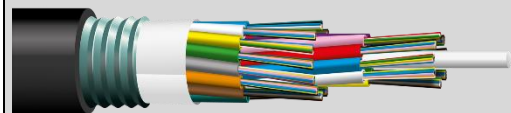


Figure 9: Fiber-Optic Cable

2. Cable Lengths:

- **General-purpose Lab 1:** Estimated 100 meters (including routing and vertical distance).
- **General-purpose Lab 2:** Estimated 120 meters.
- **Cisco Network Lab:** Estimated 110 meters.
- **IoT Lab:** Estimated 130 meters.
- **Video Conferencing Room:** Estimated 80 meters.
- **Student Lounge:** Estimated 90 meters.

Work Area	Cable Length (meters)
General-purpose Lab 1	100
General-purpose Lab 2	120
Cisco Network Lab	110
IoT Lab	130
Video Conferencing Room	80
Student Lounge	90

Table 2:

The Table shows that the estimated length of the cables.



Figure 10:

The Cables connect to the switches.

#Task 5:

Introduction

IP addressing is crucial for ensuring that every host on the network can connect without address conflicts. This task involves dividing the given network address into subnets to efficiently organize different labs and rooms within the institution.

Network Address and Subnetting

We received the network address 192.168.0.0/24 from our lecturer. This address will be subdivided to allocate IP addresses to various labs and rooms in our building. The aim is to ensure efficient use of IP addresses while accommodating future expansion.

Subnetting Strategy

Given the structure of our network and the requirement for different labs and rooms, we will employ a subnetting strategy that balances the number of hosts per subnet with the need for network separation.

1. Subnet Calculation:

- We start with the network address 192.168.0.0/24.
- This gives us 256 IP addresses (2^8).
- We need to create subnets for each lab and room, each with an appropriate number of hosts.

2. Subnet Allocation:

- **General-purpose Lab 1:**
 - Subnet: 192.168.0.0/26
 - Host range: 192.168.0.1 - 192.168.0.62
 - Broadcast address: 192.168.0.63
- **General-purpose Lab 2:**
 - Subnet: 192.168.0.64/26
 - Host range: 192.168.0.65 - 192.168.0.126
 - Broadcast address: 192.168.0.127
- **Cisco Network Lab:**

- Subnet: 192.168.0.128/27
- Host range: 192.168.0.129 - 192.168.0.158
- Broadcast address: 192.168.0.159

- **IoT Lab:**
 - Subnet: 192.168.0.160/27
 - Host range: 192.168.0.161 - 192.168.0.190
 - Broadcast address: 192.168.0.191

- **Video Conferencing Room:**
 - Subnet: 192.168.0.192/28
 - Host range: 192.168.0.193 - 192.168.0.206
 - Broadcast address: 192.168.0.207

- **Student Lounge:**
 - Subnet: 192.168.0.208/28
 - Host range: 192.168.0.209 - 192.168.0.222
 - Broadcast address: 192.168.0.223

Detailed IP Assignment

Each lab and room will be assigned specific IP addresses within their designated subnet ranges. This detailed assignment ensures that each device, such as computers, printers, and other networked equipment, has a unique IP address within the network.

1. General-purpose Lab 1:

- Computers: 192.168.0.1 - 192.168.0.30
- Printers: 192.168.0.31 - 192.168.0.32
- Network Equipment: 192.168.0.33 - 192.168.0.40
- Reserved for future use: 192.168.0.41 - 192.168.0.62

2. General-purpose Lab 2:

- Computers: 192.168.0.65 - 192.168.0.94
- Printers: 192.168.0.95 - 192.168.0.96
- Network Equipment: 192.168.0.97 - 192.168.0.104
- Reserved for future use: 192.168.0.105 - 192.168.0.126

3. Cisco Network Lab:

- Devices: 192.168.0.129 - 192.168.0.158

4. IoT Lab:

- Devices: 192.168.0.161 - 192.168.0.190

5. Video Conferencing Room:

- Devices: 192.168.0.193 - 192.168.0.206

6. Student Lounge:

- Devices: 192.168.0.209 - 192.168.0.222

Conclusion

The IP addressing scheme has been meticulously planned and documented, ensuring efficient and conflict-free connectivity for all network devices across different labs and rooms. The scheme allows for future expansion and scalability, adhering to best practices in network design.

MEETING MINUTES For Task 1

DATE/TIME		25- Mar 2024 11:00 am	
LOCATION		University Campus	
AGENDA		Project Planning and Requirements Gathering	
Meeting MC		Zhir Daniel	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Mohammed Kamal Ali		10:50	
Dina Mohammed Bilal		10:45	
Dyako Abubakr Mohammed		10:55	
Zhir Daniel		10:57	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Project Objectives	Agreed with the objectives and added the importance of keeping the budget within limits.	Zhir (03/26)
2	Requirement Gathering:	Suggested meeting with stakeholders from each lab and room to gather specific requirements.	Dyako (03/10)
3	Roles and Responsibilities	Designated as the budget and procurement manager to handle all financial aspects and purchasing of equipment.	Zhir (03/12)
4	Timeline and Milestones	Supported the timeline and stressed the importance of sticking to the schedule to avoid delays.	Mohammed (03/15)
5	Meeting ended	At 11:45 am	Next Meeting on (08/04)

MEETING MINUTES For Task 2

DATE/TIME		08 - Apr 2024 9:30 am	
LOCATION		University Campus	
AGENDA		Selection of Network Devices	
Meeting MC		Dyako Abubakr Mohammed	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Mohammed Kamal Ali		09:10	
Dina Mohammed Bilal		09:13	
Dyako Abubakr Mohammed		09:15	
Zhir Daniel		09:25	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on Network Requirements	Suggested considering future scalability when choosing devices to avoid frequent upgrades.	Dina (04/10)
2	Device Selection	Suggested ASUS routers for the General-purpose Labs due to their user-friendly interface and performance.	Dyako (04/10)
3	Evaluation of Options	Checked the warranty and support services offered by different vendors.	Zhir (04/12)
4	Final Decision	To coordinate with the IT department for installation and setup.	Mohammed (04/15)
5	Meeting ended	At 10:25 am	Next Meeting on (17/04)

MEETING MINUTES For Task 3

DATE/TIME		17 - Apr 2024 9:30 am	
LOCATION		University Campus	
AGENDA		CHOOSING THE APPROPRIATE LAN DEVICES	
Meeting MC		Mohammed Kamal Ali	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Mohammed Kamal Ali		09:17	
Dina Mohammed Bilal		09:25	
Dyako Abubakr Mohammed		09:25	
Zhir Daniel		09:23	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Router Placement and Capacity	Dina: Place routers at the main network entry points for efficient connectivity and traffic management.	Dina (04/20)
2	Determination of Network Device Placement	Dyako: Central switch placement in labs.	Dyako (04/23)
3	Calculation of Required Connections, Patch Cords, and Switch Ports	A Mohammed: Additional switch ports for expansion.	Mohammed (04/27)
4	Planning for Vertical and Horizontal Cabling Connections	Dina: Designated conduits for vertical cabling.	Dyako (04/24)
5	Meeting ended	At 10:45 am	Next Meeting on (08/05)

MEETING MINUTES For Task 4

DATE/TIME		08 - May 2024 10:30 am	
LOCATION		University Campus	
AGENDA		Making the Connections – LAN and WAN	
Meeting MC		Mohammed Kamal Ali	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Mohammed Kamal Ali		10:00	
Dina Mohammed Bilal		10:30	
Dyako Abubakr Mohammed		10:25	
Zhir Daniel			Privacy Issues
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Work Area Identification and Mapping	Dina: Color-coded labels for work areas.	Dina (05/10)
2	Determination of Network Device Placement	Dyako: Central switch placement in labs.	Dyako (05/16)
3	Calculation of Required Connections, Patch Cords, and Switch Ports	A Mohammed: Additional switch ports for expansion.	Mohammed (05/19)
4	Planning for Vertical and Horizontal Cabling Connections	Dina: Designated conduits for vertical cabling.	Dyako (05/17)
5	Meeting ended	At 11:45 am	Next Meeting on (28/05)

MEETING MINUTES For Task 5

DATE/TIME		28 - May 2024 1:00pm	
LOCATION		University Campus	
AGENDA		IP Addressing Scheme	
Meeting MC		Dina Mohammed	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Mohammed Kamal Ali		12:55	
Dina Mohammed Bilal		12:50	
Dyako Abubakr Mohammed		12:45	
Zhir Daniel		12:45	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion on Subnetting	proposed using /26 subnets for the general-purpose labs.	Dina (16/06)
2	IP Allocation	Dina allocated IP addresses for General-purpose Lab 1. Mohammed worked on General-purpose Lab 2. Dyako assigned IPs for the Cisco Network Lab and IoT Lab. Zhir handled the Video Conferencing Room and Student Lounge	Dyako (18/06)
3	Future Proofing	Discussed and agreed on reserving IP ranges for future use in each subnet.	Mohammed (17/06)
4	Documentation and Submission	Assigned roles for submitting the task: Dina will compile, and Mohammed will submit	Dyako (17/06)
5	Meeting ended	At 1:45 PM	Next Meeting on (15/06)